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Math 31 — Differential Equations
Section 7.3.1

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3. $\mathcal{L}\{t^3e^{-2t}\}$

Shift s by $(s-(-2))$:
$\mathcal{L}\{t^3e^{-2t}\} = \frac{6}{(s+2)^3}$

7. $\mathcal{L}\{e^t\sin(3t)\}$

Shift by $(s-1)$:
$\mathcal{L}\{e^t\sin(3t)\} = \frac{3}{(s-1)^2+9}$

9. $\mathcal{L}\{(1-e^t+3e^{-4t})\cos(5t)\}$

Distributing the $\cos(5t)$ into each term:
$\mathcal{L}\{\cos(5t) - e^t\cos(5t) + 3e^{-4t}\cos(5t)\}$
Which gives us:
$\mathcal{L}\{\cos(5t) - e^t\cos(5t) + 3e^{-4t}\cos(5t)\}$
Taking the Laplace:
$\frac{s}{s^2+25} - \frac{(s-1)}{(s-1)^2+25} + \frac{3s}{(s+4)^2+25}$

11. $\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^3}\right\}$

The s term here is shifted by 3 and has a multiplicity of 3:
Getting the inverse into form:
$\frac{1}{2!}\mathcal{L}^{-1}\left\{\frac{2!}{(s+3)^{2+1}}\right\}$
Which gives us our final answer of:
$\boldsymbol{\left[\frac{t^2e^{-3t}}{2}\right]}$

13. $\mathcal{L}^{-1}\left\{\frac{1}{s^2-6s+10}\right\}$

Check if the denominator is reducible:
$D = \sqrt{(-6)^2 - (4\cdot 10\cdot 1)} \rightarrow D = \sqrt{-4}$
$\therefore$ Irreducible:
Utilize completing the square:
$\mathcal{L}^{-1}\left\{\frac{1}{s^2-6s+9-9+10}\right\} \rightarrow \mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2+1}\right\}$
Which gives us our final answer of:
$\boldsymbol{[e^{3t}\sin(t)]}$

15. $\mathcal{L}^{-1}\left\{\frac{s}{s^2+4s+5}\right\}$

First complete the square on the bottom:
$\frac{s}{(s+2)^2+1}$
Take inverse which gives us our final answer:
$\boldsymbol{[e^{-2t}\cos(t)]}$

16. $\mathcal{L}^{-1}\left\{\frac{2s+5}{s^2+6s+34}\right\}$

Complete the square on the denominator:
$\mathcal{L}^{-1}\left\{\frac{2s+5}{(s+3)^2+25}\right\}$
Separate the fraction:
$\mathcal{L}^{-1}\left\{\frac{2s}{(s+3)^2+25} + \frac{5}{(s+3)^2+25}\right\}$

Which results in:

$$\left[\mathbf{2e^{-3t}\cos(5t) + 5e^{-3t}\sin(5t)}\right]$$

17. $\mathcal{L}^{-1}\left\{\frac{s}{(s+1)^2}\right\}$